MITOCHONDRIAL & CELLULAR ENERGY

MOTS-c — the full review

MITOCHONDRIAL-DERIVED PEPTIDE

This is the deeper look behind the short version — where this peptide's blueprint actually comes from, what the mouse exercise study found and what it didn't, the difference between the peptide itself and the analog that was actually tested in people, and what your consultation covers.

What it is

MOTS-c is a small, lab-synthesized chain of 16 amino acids. What makes it scientifically unusual is where its blueprint comes from: almost everything a cell makes is encoded by nuclear DNA, but MOTS-c's instructions sit inside a short stretch of the mitochondrial genome — the DNA carried by mitochondria, the structures inside cells that generate energy. For decades, mitochondrial DNA was assumed to code only for a handful of energy-machinery proteins; MOTS-c is one of a small, only recently recognized family of "mitochondrial-derived peptides" that appear to let mitochondria send signals elsewhere in the cell and body.

It was discovered in 2015 by researchers at the USC Leonard Davis School of Gerontology, through a bioinformatic screen of previously overlooked mitochondrial coding regions, published in Cell Metabolism. It is sold today as a subcutaneous injectable, through compounding pharmacies and online research-chemical vendors, marketed for metabolic health, weight loss, exercise performance, and general longevity — none of which is an FDA-recognized indication, and almost none of which has actually been tested by giving the peptide to a person.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Metabolic health and insulin sensitivity	**Animal data only**	Mouse studies show AMPK-mediated improvements in insulin sensitivity and glucose handling; in humans only correlational blood-level data exists, and the first controlled human trial only began recruiting in February 2026.
Weight loss and fat metabolism	**Animal data only**	Mouse studies report reduced weight gain and improved fat oxidation; no completed human efficacy trial of the native peptide exists, and the one related trial of a different analog found no significant difference in liver fat versus placebo.
Exercise performance	**Animal data only**	The signature finding — roughly doubled treadmill running duration in old mice — is a 2021 mouse result; the only human data shows exercise raises a person's own MOTS-c, not that injecting it improves performance.

USED FOR	EVIDENCE	WHAT THAT MEANS
Longevity and anti-aging	**Animal data only**	Based on aging-mouse studies; human data is limited to correlational studies, which does not establish that supplementing it affects human aging or lifespan.
Mitochondrial health generally	**Theoretical**	A plausible extension of MOTS-c's role as a mitochondrial-signaling molecule, but not a validated clinical endpoint tested in any human trial of this peptide.

How it works

The leading proposed mechanism is that MOTS-c activates AMPK (AMP-activated protein kinase), a master cellular fuel-sensing enzyme that also switches on with exercise, calorie restriction, and drugs like metformin. In mouse and cell studies, MOTS-c-driven AMPK activation is linked to increased glucose uptake into muscle, increased fat burning, and suppressed glucose output from the liver. A second, AMPK-independent pathway has also been proposed, in which MOTS-c moves into the cell nucleus under metabolic stress and interacts with stress-response genes.

The "exercise mimetic" reputation comes from a 2021 mouse study in which giving MOTS-c to mice, including old mice, roughly doubled treadmill running duration compared with untreated animals. The same paper found that a bout of exercise raises a person's own MOTS-c — roughly twelve-fold in skeletal muscle tissue and about 1.5 to 1.6-fold in circulating blood, in a small ten-person sample. That is a different kind of evidence than an injection working: it shows exercise triggers a rise in the body's own MOTS-c, not that injecting MOTS-c reproduces what exercise does. No human study has tested whether administering the peptide improves running capacity, VO2 max, strength, or any other performance measure.

What the research actually looked at

Two very different bodies of research exist for MOTS-c, and marketing routinely blurs them together. The first is observational human research measuring people's own natural circulating MOTS-c: several published cohort studies have linked lower blood MOTS-c to higher fasting insulin and worse insulin resistance, reported reduced levels in type 2 diabetes, and correlated levels with muscle mass and physical performance in older adults. A 2025 study of 169 people testing MOTS-c as a brain-aging biomarker found at best a weak, inconsistent signal, explicitly describing "poor discriminatory power." None of this involves giving anyone MOTS-c — it is blood-level correlation research, not evidence that adding more of it produces a benefit.

The second body of research — actually administering MOTS-c to people — is extremely thin, bordering on nonexistent for the peptide people are being sold. FDA's own review states it has "not identified any human exposure data on drug products containing MOTs-C administered via any route of administration." The only related interventional human data comes from CB4211, a chemically modified analog of MOTS-c — a different molecule, not MOTS-c itself — which went through a Phase 1a/1b randomized, placebo-controlled trial of roughly 88 people. It was reasonably well tolerated but was temporarily paused for injection-site reactions, showed modest liver-enzyme reductions but no significant difference in liver fat versus placebo, and the company developing it subsequently dissolved and discontinued the program.

For the actual native peptide, the first real controlled human efficacy trial — a Phase 2a randomized, double-blind, placebo-controlled study testing MOTS-c by subcutaneous injection in roughly 120 adults with prediabetes and overweight or obesity — only began recruiting in February 2026, with an estimated completion date of May 2028. It has reported nothing yet, which means interventional human evidence for the peptide actually sold under this name is, as of now, essentially zero completed studies.

Common misunderstandings

- **“It's an exercise-in-a-shot, proven in humans.”** The dramatic exercise-capacity finding is a mouse result. The human data runs the opposite direction: exercising raises a person's own MOTS-c — it does not show that injecting it gives you the effects of exercise.
- **“Because the body makes it naturally, it must be safe to inject at any dose.”** FDA's own conclusion is the opposite: it has not identified any human exposure data on drug products containing MOTS-c administered by any route.
- **“There's a ton of research on this, it's well established.”** The overwhelming majority is mouse and cell-culture mechanism work, plus human studies that only measure existing blood MOTS-c levels — essentially none involves administering MOTS-c to a person.
- **“The 2026 advisory vote means MOTS-c is basically approved now.”** That vote was non-binding, passed narrowly over FDA staff's explicit objection, and only concerned whether pharmacies may compound with it — not whether it works or is safe.

Who it is not for

- Anyone expecting an established human safety profile — none exists at any dose.
- Pregnant or breastfeeding patients, or anyone trying to conceive.
- Anyone with active cancer, a cancer history, or an undiagnosed mass — the mechanism touches cell-growth signaling pathways not fully characterized in humans, and no carcinogenicity studies exist.
- Anyone on metformin or another AMPK-pathway-active medication — a shared mechanism raises an unstudied interaction concern.
- Anyone sourcing product from an unverified research-use-only vendor.
- Competitive athletes subject to WADA-code testing.

Competitive athletes: MOTS-c is prohibited at all times, in and out of competition, under WADA category S4 (Hormone and Metabolic Modulators), listed as an AMPK activator. A therapeutic use exemption is not a realistic path around this, since a TUE requires an approved therapeutic use and MOTS-c has none. Anti-doping operates on strict liability regardless of a clinician's recommendation or the fact that the substance is naturally produced by the body.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, and the human record is essentially empty. FDA states it has no human exposure data for MOTS-c administered by any route. An advisory committee voted narrowly (7-5, two abstentions) in July 2026 to recommend it for a compounding list, for obesity and osteoporosis — advisory, not acted on, not approval.

The one real human trial just started. A controlled trial of the actual peptide began enrolling in February 2026 and has not reported. Everything else is mouse and cell work, or blood-level correlation studies that never administered the peptide at all.

The exercise-mimetic story is usually told backwards — the human finding is that exercise raises your own MOTS-c, not that injecting it replicates exercise.

Clinical detail

A 16-amino-acid peptide encoded within the mitochondrial genome, discovered in 2015 through a bioinformatic screen of overlooked mitochondrial coding regions, published in Cell Metabolism.

Proposed mechanism centers on AMPK activation, the same fuel-sensing pathway engaged by exercise, calorie restriction, and metformin.

FDA states it has not identified any human exposure data on drug products containing MOTS-c administered via any route of administration.

The only related interventional human data comes from CB4211, a chemically distinct analog, not native MOTS-c; its Phase 1a/1b trial found no significant difference in liver fat versus placebo, and the program was discontinued.

The first controlled human efficacy trial of the native peptide (NCT07505745, ~120 adults with prediabetes/overweight-obesity) began recruiting February 2026, with results not expected before 2028.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

MOTS-c is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.